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VICTORIA
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SCHOOL OF ECONOMICS
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ECON412 INTERNATIONAL ECONOMICS: TRADE

TRIMESTER 2, 2018

LECTURE WEEK #10: MONOPOLISTIC COMPETITION WITH HETEROGENEOUS FIRMS

2 OCTOBER 2018

OVERVIEW

- **Readings:** Feenstra Chapter 6; Melitz (2003).
- Monopolistic competition with heterogeneous firms.

1. INTRODUCTION

- In Krugman's model of increasing returns to scale and monopolistic competition
  - firms are symmetric and identical.
  - two equilibrium conditions: $MR = MC$ and $P = AC$.
- Model fails
  - to match the fact that only a small fraction of firms participating in int'l trade.
  - to capture the selection effect of trade: entry and exit induced by trade.
- Allowing firm heterogeneity can overcome the above limitation, but
  - introducing heterogeneity will violate the zero-profit condition in Krugman's models.

2. EMPIRICAL EVIDENCES ON FIRM HETEROGENEITY AND TRADE

- See works by Bernard etal (2007), and papers by Pavcnik, de Loecker, Trefler etc.
- Firms participating in trade perform better along several dimensions.
  - Strong evidence of selection into trade, some evidence of learning by exporting.
  - Firms adjust to the trading environment, in productivity, technology adoption, product types, etc.
- Large turnover of plants and firms
  - A third of U.S. manufacturing plants enter and exit every 5 years.
  - Evidence of reallocation triggered by trade liberalization.
  - Extensive margin of the number of exporting firms accounts for a large fraction of variation in exports across markets.

BERNARD ETAL (2007)

Table 3

Exporter Premia in U.S. Manufacturing, 2002

	Exporter premia		
	(1)	*(2)*	*(3)*
Log employment	1.19	0.97	
Log shipments	1.48	1.08	0.08
Log value-added per worker	0.26	0.11	0.10
Log TFP	0.02	0.03	0.05
Log wage	0.17	0.06	0.06
Log capital per worker	0.32	0.12	0.04
Log skill per worker	0.19	0.11	0.19
Additional covariates	None	Industry fixed effects	Industry fixed effects, log employment

Sources: Data are for 2002 and are from the U.S. Census of Manufactures.
Notes: All results are from bivariate ordinary least squares regressions of the firm characteristic in the first column on a dummy variable indicating firm's export status. Regressions in column 2 include industry fixed effects. Regressions in column 3 include industry fixed effects and log firm employment as controls. Total factor productivity (TFP) is computed as in Caves, Christensen, and Diewert (1982). "Capital per worker" refers to capital stock per worker. "Skill per worker" is nonproduction workers per total employment. All results are significant at the 1 percent level.

BERNARD ETAL (2007)

Table 5

The Intensive and Extensive Margins of Exporters, 1997

	Exporter premia	
	(1)	*(2)*
Log number of products	0.23	0.27
Log mean shipments/# products	1.25	0.73
Additional covariates	None	Industry fixed effects

Sources: Data are for 1997 and are from the U.S. Census of Manufactures.
Notes: All results are from bivariate ordinary least squares regressions of the firm characteristic in the first column on a dummy variable indicating firm export status. Regressions in column two include four-digit SIC industry fixed effects. The first dependent variable is the log of the number of five-digit SIC products produced by the firm in 1997. The second dependent variable is the log of total firm shipments divided by the number of products.

3. MELITZ MODEL

- Original model allows for many sectors in the economy, we assume that the economy has only **one sector** (or no sector).
- Starting with one economy, equilibrium in autarky, then equilibrium with trade.

3-1. PREFERENCE AND DEMAND

- There are L households, each supplying one unit of labor.
- CES preference for a continuum of differentiated products in a sector, $Q = \left[\int_\Omega q(\omega)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$ with $\sigma > 1$.
- Each consumer earns wage w. Demand for product ω is $q(\omega) = \left[\frac{p(\omega)}{P} \right]^{-\sigma} \frac{w}{P}$, where aggregate price $P = \left[\int_\Omega p(\omega)^{1-\sigma} d\omega \right]^{\frac{1}{1-\sigma}}$.
- Price elasticity of demand for each product: $-\ln q(\omega) / \ln p(\omega) = \sigma$.

3-2. FIRMS

- Labor requirement $l = f_d + \frac{y}{\varphi}$, if output is sold in domestic market.
- There is a continuum of firms with varying productivity φ. Let $g(\varphi)$ and $G(\varphi)$ be respectively pdf and cdf.
- Each firm produces a differentiated product (variety).

3-3. EQUILIBRIUM IN CLOSED ECONOMY

- Marginal cost is w/φ. Optimal pricing by $MR = MC$ suggests $p(\omega) = \frac{\sigma}{\sigma-1} \cdot \frac{w}{\varphi}$. Here after, we use φ to index products.

- Total demand is $Lq(\varphi) = Lp(\varphi)^{-\sigma} w / P_a^{1-\sigma}$, with P_a price index in autarky. Output market clearing, $y(\varphi) = Lq(\varphi)$. Substitute for $p(\varphi)$, output is given by
$$y(\varphi) = \frac{L}{P_a^{1-\sigma}} \left[\frac{\sigma}{\varphi(\sigma-1)}\right]^{-\sigma} w^{1-\sigma}.$$

- Firm profit is $\pi(\varphi) = p(\varphi)y(\varphi) - w[f_d + y(\varphi)/\varphi] = \frac{wy(\varphi)}{\varphi(\sigma-1)} - wf_d$. Substitute for $y(\varphi)$, we have
$$\pi(\varphi) = \frac{wL\sigma^{-\sigma}}{P_a^{1-\sigma}(\sigma-1)^{1-\sigma}} w^{1-\sigma}\varphi^{\sigma-1} - wf_d = B_a\varphi^{\sigma-1} - wf_d.$$
  - Profits increase with productivity φ and decrease with fixed cost f_d.

- **Zero-cutoff-profit** (ZCP) condition: $\pi(\varphi) = 0$. There is a $\varphi = \varphi_a$ at which profits are zero.

$$\varphi_a^{\sigma-1} = \frac{f_d \sigma^\sigma w^{\sigma-1}}{L P_a^{\sigma-1} (\sigma - 1)^{\sigma-1}} = w f_d / B_a.$$

- Firms with a $\varphi \geq \varphi_a$ **survive** and earn non-negative profits. Firms with $\varphi < \varphi_a$ do not produce, they **exit** the market.

- Different from Krugman's model, here the ZCP condition does not pin down φ_a.

  - φ_a depends on price index P_a in autarky, which in turn depends on the range of variety produced.
  - φ_a is pinned down by the free entry condition.

- Equilibrium requires the *free entry* condition: the expected profits from entering the market must be zero in equilibrium.

- **Entry**. Let f_e be the fixed entry cost (measured in units of labor). Productivity φ is drawn from $g(\varphi)$ *only* having paid the entry cost. φ is fixed at the value drawn at entry.

- **Free entry** implies that in equilibrium, expected ex-ante profits equal to zero: $\int_0^\infty \pi(\varphi)g(\varphi)d\varphi - wf_e = 0$.

- Re-write profit in terms of φ and φ_a, noting $B_a = wf_d/\varphi_a^{\sigma-1}$, then $\pi(\varphi) = \left[\left(\varphi/\varphi_a\right)^{\sigma-1} - 1\right]wf_d$. Free-entry condition becomes
$$wf_e = \int_0^\infty \pi(\varphi)g(\varphi)d\varphi = wf_d \int_{\varphi_a}^\infty \left[\left(\varphi/\varphi_a\right)^{\sigma-1} - 1\right]g(\varphi)d\varphi.$$

- Assume that $\varphi \in [1, \infty)$ and follows a Pareto distribution: $G(\varphi) = 1 - \varphi^{-\theta}$. Then $g(\varphi) = \theta\varphi^{-\theta-1}$.
  - Restrict $\theta > \sigma - 1$, so that trade flows are finite (later).
  - Pareto distribution is proposed not by Melitz (2003), but by Chaney (2008), to derive the gravity equation.
- Free-entry condition then becomes $f_e = J(\varphi_a) f_d$, with $J(\varphi_a) = \frac{\sigma-1}{\theta-(\sigma-1)} \varphi_a^{-\theta} = \frac{\sigma-1}{\theta-(\sigma-1)} [1 - G(\varphi_a)]$.
- $J(\varphi_a)$ decreases as φ_a increases. We solve for the cutoff value in equilibrium, $\varphi_a = \left[\frac{\sigma-1}{\theta-(\sigma-1)} \cdot \frac{f_d}{f_e}\right]^{1/\theta}$.
  - A lower fixed production cost and a higher entry cost imply a lower φ_a.

- Firm selection (φ_a) is only determined by demand elasticity and fixed costs
  - Aggregate demand and input supply does not matter, a result owing to the CES preference and monopolistic competition.
- Finally, need to obtain the **mass of active firms** (products) in equilibrium.
- **Expected profit conditional on drawing** a $\varphi > \varphi_a$ at entry: $\overline{\pi}(\varphi_a) = \int_{\varphi_a}^{\infty} \pi(\varphi) \frac{g(\varphi)}{1-G(\varphi_a)} d\varphi = \frac{wf_e}{1-G(\varphi_a)}$, where $\frac{g(\varphi)}{1-G(\varphi_a)}$ is the density function of **active** firms (they have $\varphi > \varphi_a$). We have used the free-entry condition in the second equality.
- Let M_a be the equilibrium **mass** of active firms or products in autarky (equivalent to the number of varieties in the discrete case).
- Suppose the mass of entering firms paying the sunk entry cost is M_e, only a fraction $[1 - G(\varphi_a)]$ obtain a successful draw of $\varphi > \varphi_a$. Then $M_a = M_e[1 - G(\varphi_a)]$.

- The **full-employment** condition pins down M_a.

$$
\begin{aligned}
L &= M_a \int_{\varphi_a}^{\infty} \left[f_d + \frac{y(\varphi)}{\varphi} \right] \frac{g(\varphi)}{1 - G(\varphi_a)} d\varphi + M_e f_e \\
&= M_a \int_{\varphi_a}^{\infty} \left[(\sigma - 1)\pi(\varphi)/w + \sigma f_d \right] \frac{g(\varphi)}{1 - G(\varphi_a)} d\varphi + \frac{M_a f_e}{1 - G(\varphi_a)} \\
&= \sigma M_a [\overline{\pi}(\varphi_a)/w + f_d].
\end{aligned}
$$

  Here we have used $\pi(\varphi) = \frac{wy(\varphi)}{\varphi(\sigma-1)} - wf_d$ and the conditional expected profits $\overline{\pi}(\varphi_a)$.

- Thus, the equilibrium mass of products is $M_a = \frac{L}{\sigma[\overline{\pi}(\varphi_a)/w + f_d]} = \frac{L/\sigma}{\frac{f_e}{1-G(\varphi_a)} + f_d}$.

- We have obtained **equilibrium in closed economy**: φ_a, M_a. Price index is then $P_a = \left[M_a \int_{\varphi_a}^{\infty} p(\varphi)^{1-\sigma} \frac{g(\varphi)}{1-G(\varphi_a)} d\varphi \right]^{1/(1-\sigma)}$.

3-4. EQUILIBRIUM WITH TRADE

- Trade between two identical countries, Home and Foreign.
- Assume a fixed cost f_x to participate in exporting. Think of learning foreign markets, setting up distribution networks, etc.
- Assume an iceberg trade cost $\tau > 1$, selling one unit abroad requires producing τ units.
- Further, normalize wages to be one in both countries.
- Domesetic price, $p_d(\varphi) = \frac{\sigma}{\varphi(\sigma-1)}$; Export price, $p_x(\varphi) = \frac{\sigma\tau}{\varphi(\sigma-1)}$.
- **ZCP condition in domestic market**: $\pi_d(\varphi_d) = B_d\varphi_d^{\sigma-1} - f_d = 0$, or $\varphi_d^{\sigma-1} = \frac{f_d}{B_d} = \frac{f_d\sigma^\sigma}{LP_d^{\sigma-1}(\sigma-1)^{\sigma-1}}$.

- Foreign demand is $y_x(\varphi) = Lp_x(\varphi)^{-\sigma}/P_d^{1-\sigma}$, where P_d is price index in Foreign country which is the same as price index at Home by symmetry.

- Foreign demand is then $y_x(\varphi) = \frac{L}{P_d^{1-\sigma}}\left[\frac{\sigma\tau}{\varphi(\sigma-1)}\right]^{-\sigma}$.

- Profits from exporting, $\pi_x(\varphi) = p_x(\varphi)y_x(\varphi) - (f_x + \frac{\tau y_x(\varphi)}{\varphi}) = \frac{\tau y_x(\varphi)}{\varphi(\sigma-1)} - f_x$. Substitute for $y_x(\varphi)$, we have
$$\pi_x(\varphi) = \frac{L\tau^{1-\sigma}\sigma^{-\sigma}}{P_d^{1-\sigma}(\sigma-1)^{1-\sigma}}\varphi^{\sigma-1} - f_x = B_x\varphi^{\sigma-1} - f_x.$$

- **ZCP condition in the export market** is $\pi_x(\varphi_x) = B_x\varphi_x^{\sigma-1} - f_x = 0$, or $\varphi_x^{\sigma-1} = f_x/B_x = \frac{f_x\tau^{\sigma-1}\sigma^{\sigma}}{LP_d^{\sigma-1}(\sigma-1)^{\sigma-1}}$.

- From ZCP conditions in domestic and export markets, we obtain $\left(\frac{\varphi_x}{\varphi_d}\right)^{\sigma-1} = \frac{f_x \tau^{\sigma-1}}{f_d}$.
  - We impose that $f_x \tau^{\sigma-1} > f_d$: only a subset of active firms sell in both markets.
  - Then, $\varphi_x > \varphi_d$, only more productive firms export.
- To pin down φ_x and φ_d, we again use the free-entry conditions: $\int_0^\infty [\pi_d(\varphi) + \pi_x(\varphi)] g(\varphi) d\varphi = f_e$.
- After some steps, we can show that $J(\varphi_d) f_d + J(\varphi_x) f_x = f_e$, where $J(\varphi_d) = \int_{\varphi_d}^\infty \left[\left(\varphi/\varphi_d\right)^{\sigma-1} - 1\right] g(\varphi) d\varphi$ and $J(\varphi_x) = \int_{\varphi_x}^\infty \left[\left(\varphi/\varphi_x\right)^{\sigma-1} - 1\right] g(\varphi) d\varphi$. Note $J(\varphi_d)$ decreases as φ_d increases, similar for $J(\varphi_x)$.
- Comparing free-entry conditions with trade and in autarky, we must have $\varphi_a < \varphi_d < \varphi_x$.
  - Trade triggers exit.
- This implies that aggregate price is lower under trade than in autarky, $P_d < P_a$.

4. GAINS FROM TRADE

- See Feenstra (2016) book pp 164-166 for details.
- Assume that $G(\varphi) = 1 - \varphi^{-\theta}$, as in Chaney (2008).
- Let λ_d be the share of expenditure on domestic goods in total expenditure.
- It can be shown that $\lambda_d^{-1/\theta} = \varphi_d / \varphi_a$.
- Normalizing wages to one, the welfare gain from autarky to free trade equals
$$\left(\frac{1/P_d}{1/P_a}\right) = \lambda_d^{-1/\theta}.$$
- Welfare gain arises from the *selection* effect: firms with low productivity exit when the economy is open to trade.
- Can also derive the gravity equation from the Melitz-Chaney model (skipped).